CompTIA N+ Assessment - 2

Network Security, Maintenance & Troubleshooting Procedures

Section A: Concept Application

S1. Choosing CPU, RAM and Storage

Question: What factors should be considered when selecting the CPU, RAM, and storage for good performance and hardware compatibility?

The employees will run programming software, web browsers and video conferencing apps at the same time, so the computer must handle many tasks together. These are the main things I would check.

- CPU: Choose a CPU with more cores and threads (for example Intel Core i5/i7 or AMD Ryzen 5/7) because many apps run together. A good clock speed also helps compiling code and video calls. The CPU socket must match the motherboard socket (for example LGA1700 or AM5), otherwise it will not fit.
- RAM: Programming tools, browsers with many tabs and video calls use a lot of memory. I would take at least 16 GB, and 32 GB if the budget allows. The RAM type (DDR4 or DDR5) must match what the motherboard supports, and the speed should be supported too. Dual-channel (two sticks) gives better performance.
- Storage: An SSD (preferably NVMe) is much faster than an HDD, so the system boots fast and projects open quickly. 512 GB is a good minimum. The motherboard must have the right slot (M.2 or SATA) for the drive.
- Compatibility: CPU socket, RAM type, storage interface and PSU wattage should all match the motherboard. Cooling and the power supply must also be enough for the CPU. I would check the motherboard manual or the manufacturer's compatibility list before buying.

S2. Antivirus Slowing Down the Computers

Question: Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

Antivirus runs in the background all the time to protect the system, so it uses system resources.

- CPU: It scans files, programs and downloads in real time. Full system scans and scheduled scans use a lot of CPU, so other programs get less processing power and feel slow.
- RAM: The antivirus service and its protection modules stay loaded in memory. On computers with little RAM, this leaves less memory for other apps, and the system starts using the slower page file.
- Storage: Whenever a file is opened, copied or saved, the antivirus checks it. During a full scan it reads almost the whole disk, which creates heavy disk activity. This is more noticeable on an HDD than on an SSD.

Because all three are used at the same time, the computer becomes slow, especially at startup and while opening files. To reduce this, I can schedule scans after office hours, add safe folders to exclusions and make sure the computer has enough RAM and an SSD.

S3. USB Keyboard and Printer Not Working on One Computer

Question: Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

1. Check the basics first: try a different USB port, a different cable, and plug the same devices into the working computer again to confirm the devices are fine.
2. Open Device Manager and look for a yellow warning mark, "Unknown device" or a USB error on the problem computer.
3. Check the Windows version and compare it with the working computer. Check whether the printer driver supports this OS (32-bit or 64-bit also matters).
4. Uninstall the device from Device Manager, unplug it, restart the computer and plug it in again so Windows can detect it fresh.
5. Update or reinstall the driver. I would download the correct driver from the manufacturer's website instead of using an old driver.
6. If the driver is new and the problem started after an update, use "Roll Back Driver" or System Restore.
7. Run Windows Update to install missing OS patches, and use the Windows Troubleshooter for the printer and keyboard.
8. Check that the Print Spooler service is running and check USB Root Hub settings (power saving can switch off USB ports).
9. If nothing works, check the BIOS for disabled USB ports, and as a last option repair or reinstall the OS.

S4. Power Cuts and Voltage Fluctuations

Question: Explain how a UPS and surge protector help protect computer hardware and prevent data loss.

- UPS (Uninterruptible Power Supply): It has a battery inside. When the power goes out suddenly, the computer keeps running on the battery for a few minutes. This gives time to save work and shut down properly, so files do not get corrupted and the hard disk is not damaged. Many UPS units also have AVR, which keeps the voltage steady when it goes up and down.
- Surge protector: It blocks sudden high-voltage spikes (for example from lightning or when the power comes back) and sends the extra voltage to the ground. This protects the motherboard, PSU and other parts from burning or getting damaged.

Together, the surge protector protects the hardware from spikes and the UPS protects the data and work from sudden power loss. Using both is a good practice for an office that has frequent power problems.

S5. DNS and DHCP Problem

Question: Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured.

DHCP (Dynamic Host Configuration Protocol) gives network settings to a device automatically when it connects. It gives the IP address, subnet mask, default gateway and also the DNS server address. DNS (Domain Name System) then changes names (like appserver.nexus.local) into IP addresses so people do not need to remember numbers.

So they work together like this: a device connects, DHCP gives it an IP address and the DNS server information, and later the device uses that DNS server to find systems by name.
Symptoms in this scenario:

- Reaching the servers by IP address works, so the basic network connection is fine.
- Reaching them by name fails, which points to a DNS problem (wrong or missing DNS server address, or missing DNS records).
- Devices sometimes do not get a valid IP address, which points to a DHCP problem (for example the address pool is full or the scope is wrong).

Conclusion: Both services have an issue, but the main misconfiguration is in DHCP, because it is the one that fails to give a valid IP and it also hands out the DNS information. If the DHCP scope does not include the correct DNS server, name resolution fails as well. I would fix the DHCP scope first (enough addresses, correct gateway and DNS server), then check the DNS records. I can test with ipconfig /all, nslookup and ping.

S6. Firewall Blocking the Internal Application Server

Question: Describe the troubleshooting steps to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

1. Confirm the problem: internet works, but the application server is not reachable. Check if it is for all users or only some.
2. Check the basics: ping the server IP, check the server is running, and check IP/subnet/gateway on the client. Use telnet or Test-NetConnection to test the exact port of the application.
3. Log in to the firewall and look at its logs. If the logs show "deny" for traffic going to the server's IP and port, then the firewall is blocking it.
4. Check the firewall rules: look at the source, destination, port, protocol and the order of the rules. A rule placed above can block the traffic before the allow rule is reached. Also check if the new firewall has a default "deny all" rule for the internal zone.
5. Check if the server is in a different zone or VLAN and whether routing/NAT is correct between the zones.

6. Fix the issue: create or edit an allow rule only for the needed source, the server IP and the required port. Do not open everything, because that is not safe.
7. Test again from a client, check the logs for allowed traffic, and document the change.

Section B: Practical Application

Task 1: Workstation Assembly

Tools used: PC tool kit (screwdriver, anti-static wrist strap), thermal paste, and the PC components. I worked on a clean table and wore an anti-static strap to avoid damage from static electricity.

Components used: CPU, CPU cooler, motherboard, RAM, SSD, PSU, cabinet (chassis) and front panel cables.

Installation sequence:

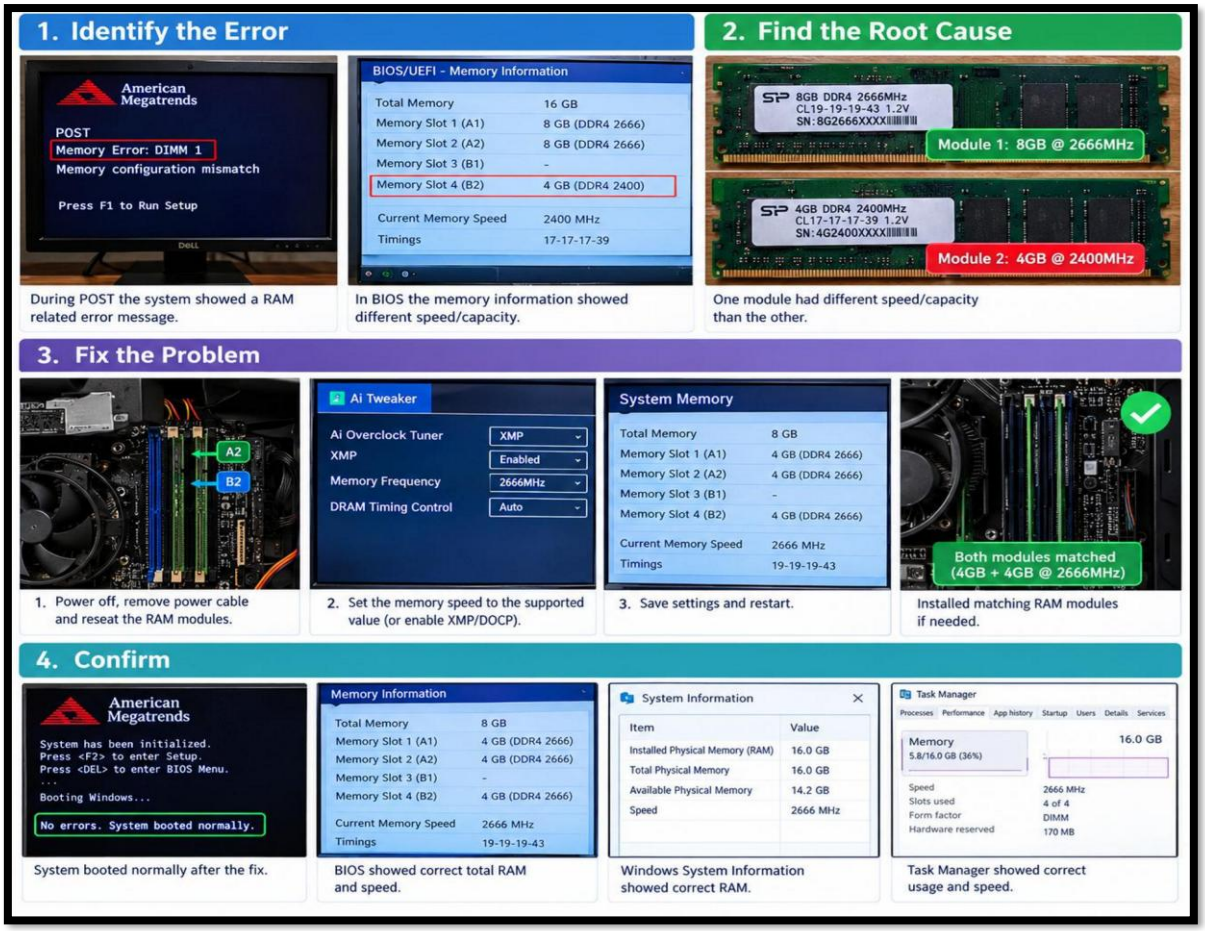
1. Opened the cabinet and kept all components and tools ready.
2. Placed the motherboard on its box and installed the CPU in the socket. I matched the triangle mark on the CPU with the socket and closed the lever.
3. Applied a small amount of thermal paste and fixed the CPU cooler, then connected the cooler fan cable to the CPU_FAN header.
4. Installed the RAM in the correct slots (pushed until both clips clicked).
5. Fixed the motherboard I/O shield in the cabinet, placed the motherboard on the standoffs and screwed it in.
6. Installed the SSD in the drive bay (or M.2 slot) and connected the SATA data cable if needed.
7. Installed the PSU in the cabinet.
8. Connected the internal cables: 24-pin motherboard power, 8-pin (4+4) CPU power, SATA power for the drive, and the SATA data cable.
9. Connected the front panel cables (power switch, reset switch, HDD LED, power LED), front USB and audio headers, as per the motherboard manual.
10. Did cable management and checked that no cable touches the fans.
11. Connected the monitor, keyboard and power cable and turned on the system.

Result: The system powered on, the fans were running and the BIOS/UEFI screen appeared.
The system passed POST (one short beep / no error), which means the assembly was correct.


1
msi
B550M
PRO-VDH WIFI
650W
2
RYZEN
3
AMD
CPU_FAN
4
XPG
5
msi
6
SAMSUNG
M.2
7
650W
8
9
PWR_SW
RESET
F_PANEL
USB
HD AUDIO
10
11
msi CLICK BIOS 5
CPU Speed 3.60 GHz
DDR Speed 2666 MHz
CPU Temperature: 36°C
Motherboard Temperature: 32°C
VCore: 1.072V
BIOS Version: E7B85IMS.1A0
BIOS Build Date: 07/15/2024
EZ Mode
CPU
Memory
Storage
Fan Info
Help
12

Task 2: RAM Mismatch Troubleshooting

1. Identify the error: During POST the system showed a RAM-related message / beep code, or the BIOS showed a different RAM speed or capacity than expected. I checked the BIOS/UEFI memory information page.
2. Find the root cause: I compared the two RAM modules. In the BIOS and on the sticker of the modules, I checked speed (MHz), capacity (GB) and timings. One module had a different speed/capacity from the other, so the system could not run both properly.
3. Fix the problem: I switched off the system, removed the power cable and reseated the RAM modules in the correct slots (A2 and B2 as per the manual). In the BIOS I set the memory to the speed supported by both modules (or enabled the correct XMP/DOCP profile). If the modules still did not work together, I used matching RAM modules.
4. Confirm: I saved the settings, restarted, and the system booted normally. The BIOS and the OS (System Information / Task Manager) showed the correct total RAM and speed.



Task 3: LAN Topology Build (Cisco Packet Tracer)

Network used: 192.168.1.0/24 (Class C), subnet mask 255.255.255.0.

Devices: 1 switch (Cisco 2960), 5 PCs and 1 network printer. All devices are connected to the switch with copper straight-through cables.

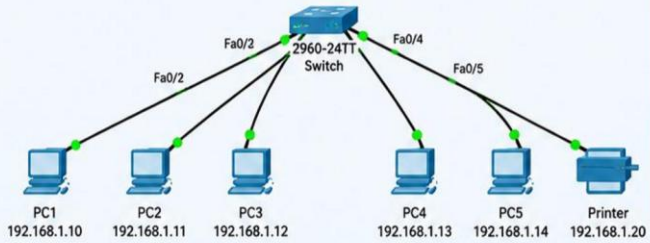
Device	IP Address	Subnet Mask	Connected to
PC1	192.168.1.10	255.255.255.0	Switch Fa0/1
PC2	192.168.1.11	255.255.255.0	Switch Fa0/2
PC3	192.168.1.12	255.255.255.0	Switch Fa0/3
PC4	192.168.1.13	255.255.255.0	Switch Fa0/4
Device	**IP Address**	**Subnet Mask**	**Connected to**
PC5	192.168.1.14	255.255.255.0	Switch Fa0/5
Printer	192.168.1.20	255.255.255.0	Switch Fa0/6

Steps:

1. Opened Packet Tracer and placed one 2960 switch, 5 PCs and 1 printer on the workspace.
2. Connected each device to the switch using copper straight-through cable.
3. Clicked each device, opened the IP Configuration (Desktop tab) and set the static IP address and subnet mask from the table above.
4. Waited for all link lights to turn green.
5. Tested connectivity by opening Command Prompt on PC1 and pinging every other device (for example: ping 192.168.1.14 and ping 192.168.1.20).

Result: All pings were successful (replies received, 0% loss), so all devices are reachable.

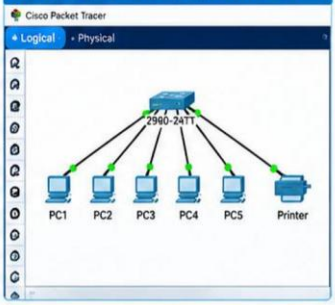
1. Final Network Topology



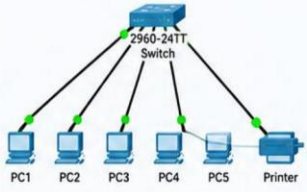
IP Address Table

Device	IP Address	Subnet Mask	Connected to
PC1	192.168.1.10	255.255.255.0	Switch Fa0/1
PC2	192.168.1.11	255.255.255.0	Switch Fa0/2
PC3	192.168.1.12	255.255.255.0	Switch Fa0/3
PC4	192.168.1.13	255.255.255.0	Switch Fa0/4
PC5	192.168.1.14	255.255.255.0	Switch Fa0/5
Printer	192.168.1.20	255.255.255.0	Switch Fa0/6

2. Devices Placed in Packet Tracer

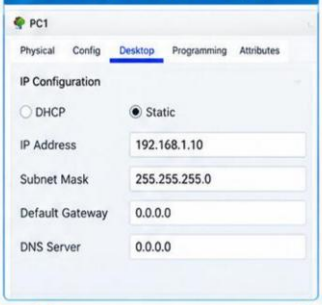


3. Connected with Copper Straight-Through



Connection type: Copper Straight-Through

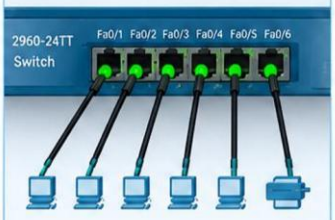
4. IP Configuration (PC1 Example)



5. Set IP for All Devices

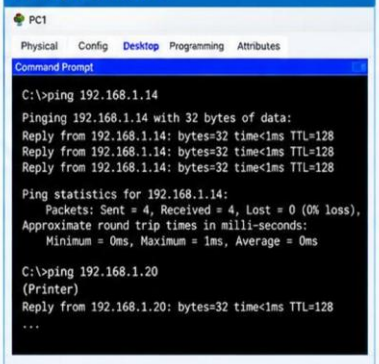
PC1	192.168.1.10 /24
PC2	192.168.1.11 /24
PC3	192.168.1.12 /24
PC4	192.168.1.13 /24
PC5	192.168.1.14 /24
Printer	192.168.1.20 /24

6. Link Lights Green



All link lights are green

7. Ping Test from PC1



8. All Devices Reachable

- PC1 → 192.168.1.11 (Success)
- PC1 → 192.168.1.12 (Success)
- PC1 → 192.168.1.13 (Success)
- PC1 → 192.168.1.14 (Success)
- PC1 → 192.168.1.20 (Success)

Result:
All pings successful (0% loss).
All devices are reachable.

Network Summary

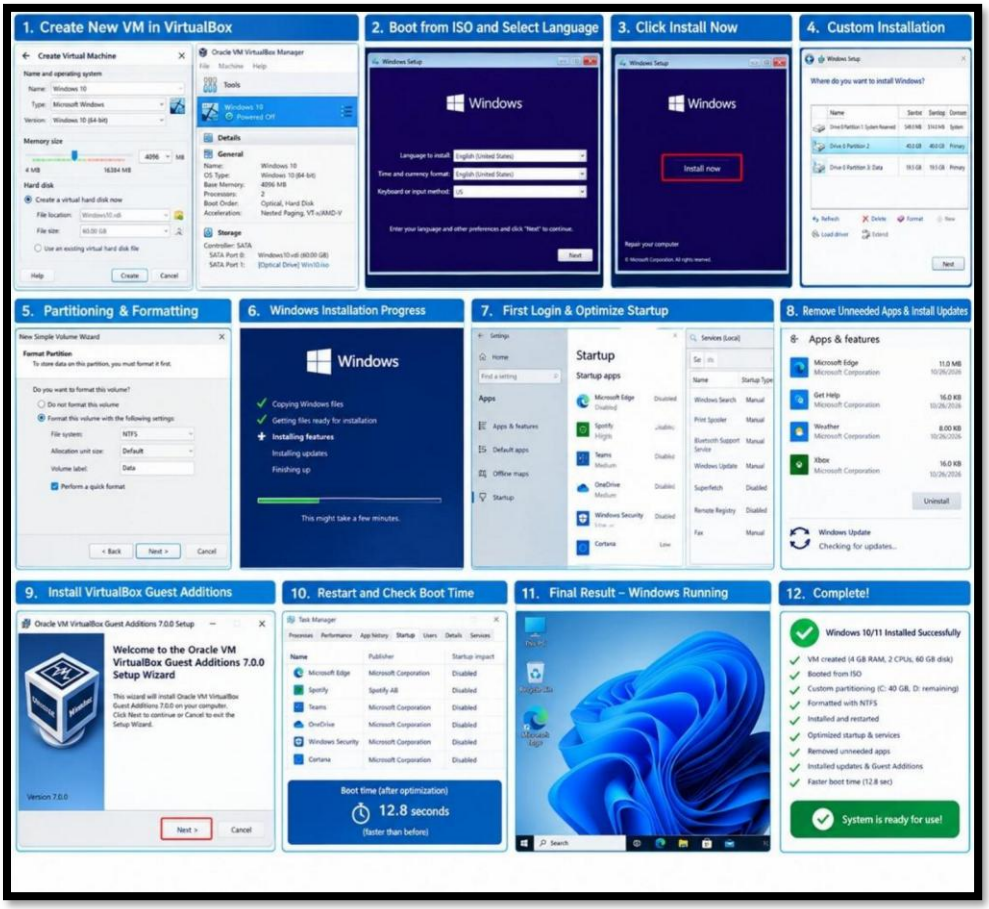
- **Switch:** Cisco 2960
- **PCs:** 5 (Static IP)
- **Printer:** 1 (Static IP)
- **Network:** 192.168.1.0/24
- **Subnet Mask:** 255.255.255.0
- **Cable:** Copper Straight-Through

Task 4: Clean OS Installation

OS used: Windows 10/11 installed in VirtualBox from an ISO file (Linux can be done in the same way).

1. Created a new virtual machine in VirtualBox, gave it 4 GB RAM, 2 CPUs and a 60 GB virtual disk, and attached the Windows ISO.
2. Started the VM and booted from the ISO. Selected language and clicked Install Now.
3. Selected "Custom: Install Windows only (advanced)" so I could do the partitioning myself instead of automatic.
4. Manually created the partitions: Partition 1 for the OS (C: drive, about 40 GB) and Partition 2 for data (D: drive, remaining space). Windows also created its small system/boot partitions.
5. Formatted the partitions with NTFS and installed Windows on the OS partition. The system restarted a few times and completed the setup.
6. After the first login, I optimized the startup: opened Task Manager > Startup and disabled unnecessary programs, and opened services.msc to set rarely used services to Manual.
7. Also removed unneeded apps, installed updates and drivers (VirtualBox Guest Additions).

Restarted the system and checked the boot time in Task Manager (Startup impact). The OS booted faster and without any errors.


1. Create New VM in VirtualBox
2. Boot from ISO and Select Language
3. Click Install Now
4. Custom Installation
5. Partitioning & Formatting
6. Windows Installation Progress
7. First Login & Optimize Startup
8. Remove Unneeded Apps & Install Updates
9. Install VirtualBox Guest Additions
10. Restart and Check Boot Time
11. Final Result – Windows Running
12. Complete!
Windows 10/11 Installed Successfully
System is ready for use!

Section C: Mini Capstone Project

Project: Nexus Innovations, Diagnosing Random System Reboots

1. Three Components Most Likely to Cause Random Reboots

- CPU / CPU cooler: When the cooler or airflow is poor, the CPU gets too hot under heavy load. At a very high temperature the CPU protects itself by shutting down or restarting the system.
- PSU (Power Supply Unit): Rendering uses CPU and GPU at full power. If the PSU is weak, old or dusty, the voltage drops under load and the computer restarts suddenly.
- GPU: Rendering also pushes the graphics card hard. An overheating GPU or one that needs more power than the PSU can give can cause a crash or restart.

(The motherboard VRM and RAM can also be affected by heat, but these three are the most likely.)

2. Why They Pass POST in the Morning but Fail in the Afternoon

In the morning the computers are cold and there is no load. POST only checks basic hardware, which uses very little power and creates very little heat, so it passes. In the afternoon the heavy rendering work starts. The CPU, GPU and PSU work at full power and produce a lot of heat. Because the computers are kept inside closed, poorly ventilated cabinets, the hot air stays inside and the temperature keeps rising during the day. After some time, the parts reach a critical temperature or the PSU cannot handle the load, and the system restarts. Dust inside the computer also makes it worse. So the problem appears only when heat and load are high.

3. Using a Stress-Test Tool to Confirm the Issue

1. Install a monitoring tool like HWMonitor or HWiNFO and a stress-test tool like Prime95, OCCT or FurMark (for the GPU).
2. Note the idle temperatures and voltages first.
3. Run the stress test for 20 to 30 minutes with the computer kept inside the cabinet, like in the real situation.
4. Watch the CPU and GPU temperatures and the +12V, +5V and +3.3V voltage readings. If the temperature goes above about 90 °C, or if the CPU throttles or the system restarts, then it is a thermal problem.

5. If the system restarts while temperatures are normal and the voltage drops a lot, then it is more likely a power (PSU) issue.
6. Repeat the test after fixing the problem to compare the results.

4. Three Steps to Fix the Restarts

1. Improve ventilation and airflow: Take the computers out of the closed cabinets, or open the cabinet back and add cooling fans. Keep some space around the computer so hot air can escape.
2. Clean and re-cool: Clean the dust from fans, heatsinks and filters with compressed air. Replace the old thermal paste on the CPU and check that all fans spin properly. Add extra case fans if needed.
3. Check and test the power: Test the PSU under load, check the cables and connections, and connect the computers to a good UPS/surge protector. Replace the PSU if it fails the test or is too weak.

5. Is a PSU Upgrade Necessary?

It depends on the test results. If the stress test shows that temperatures are high, the main problem is heat, so better cooling will solve it and a new PSU is not needed. The PSU upgrade is needed if the voltage drops under load, if the PSU is old, or if its wattage is lower than what the CPU and GPU need together. For rendering workstations, the PSU should have about 20 to 30 percent more wattage than the total system power, with a good efficiency rating (80 Plus Bronze or higher). A PSU that works near its limit also produces more heat and gives unstable power. So I would first fix the cooling and test; if the restarts continue, I would upgrade the PSU.

Section D: AI-Augmented Learning

Item 1: Exact Prompt Given to the AI Tool

Prompt:

> **Create an IP addressing and Subnetting plan for the startup office network of Nexus Innovations. The office has 5 computers (systems) and 1 network printer. Use a private Class C network. The plan must clearly show the subnet, the subnet mask, the usable host range and the broadcast address. Allocate an IP address to each of the 5 systems and the printer. Also tell me which addresses should be reserved for future expansion. Show the answer in a simple table.**

Item 2: AI's Original Plan

Item	AI's original plan
Network address	192.168.10.0
Subnet mask	255.255.255.0 (/24)
Usable host range	192.168.10.1 to 192.168.10.254
Broadcast address	192.168.10.255
Systems (PC1 to PC5)	192.168.10.1 to 192.168.10.5
Printer	192.168.10.6
Reserved for expansion	192.168.10.7 to 192.168.10.10

Item 2 (continued): My Corrected Version

Item	Corrected plan
Network address	192.168.10.0
Subnet mask	255.255.255.224 (/27)
Usable host range	192.168.10.1 to 192.168.10.30 (30 hosts)
Broadcast address	192.168.10.31
Gateway (router)	192.168.10.1
Systems (PC1 to PC5)	192.168.10.2 to 192.168.10.6
Printer	192.168.10.7
Network devices (switch/AP)	192.168.10.8 to 192.168.10.10
Reserved for expansion	192.168.10.11 to 192.168.10.30 (20 addresses)

Check I did by hand: A /27 has 32 addresses in each block (2^5 = 32). The blocks start at .0,

.32, .64 and so on. So the network address is .0 and the broadcast is .31. Usable hosts = 32 - 2 = 30. The 5 systems + 1 printer + 1 gateway + 3 network devices = 10 addresses, and 20 addresses are still free for growth. No ranges overlap.

Item 3: Note on What I Changed and Why

The AI used .1 for the first PC but did not keep any address for the gateway/router, so the network could not reach outside. I moved the gateway to .1 and shifted the systems to .2-.6 and the printer to .7. The AI also used a /24, which gives 254 hosts for only 6 devices and wastes addresses, so I changed it to a /27 (mask 255.255.255.224, broadcast .31). Its expansion space was only 4 addresses (.7-.10) and it did not leave room for the switch or access point, so I reserved .8-.10 for network devices and .11-.30 for future growth. After the changes there are no overlaps, the network and broadcast addresses are correct and there is enough space to expand.